White Paper 9 — Building MD-ready systems for the cluster

Pipeline step: Build MD-ready systems for the cluster **Inputs:** RORγt receptor, top-hit and crystal-ligand poses **Outputs:** `psoriasis_md_package.tar.gz` (equilibrated systems, production + analysis scripts)

1. Purpose

Docking and co-folding are static snapshots. Molecular dynamics (MD) simulates the protein–ligand complex in explicit water over time, testing whether a predicted binding pose is *stable* and revealing the induced-fit and solvent effects that static methods miss. This step builds fully parameterized, solvated, and equilibrated MD systems and packages them for production on the user's GPU cluster.

2. The engine: OpenMM

Systems were built and equilibrated with **OpenMM**, a high-performance, GPU-accelerated MD toolkit with a Python interface that makes system construction scriptable and reproducible [1]. OpenMM runs the same input on CPU (used here for build/equilibration) and CUDA (used on the cluster for production), so the shipped systems are directly portable.

3. Force fields: protein, ligand, water

An MD system is only as good as its force field. We used a standard, well-validated combination:

- **Protein — Amber ff14SB.** The ff14SB force field refines backbone and side-chain torsions to reproduce experimental conformational preferences and is the community standard for protein MD [2]. It sits within the broader Amber biomolecular simulation framework [3].
- **Ligand — OpenFF 2 (Sage).** Small-molecule parameters came from the **Open Force Field 2.0 (Sage)** line, a systematically-benchmarked, direct-chemical-perception force field [4]. Its lineage traces to the **General Amber Force Field (GAFF)** approach to covering arbitrary organic chemistry [5].
- **Water — TIP3P.** Explicit solvent used the TIP3P three-site water model, the standard partner for Amber protein force fields [6].

4. System construction

For each complex the pipeline: (1) fixed the receptor (missing atoms/residues, protonation at pH 7.4); (2) parameterized the ligand with OpenFF; (3) combined protein + ligand; (4) solvated in a TIP3P box with 1.0 nm padding; (5) neutralized and added 0.15 M NaCl (physiological ionic strength). The RORγt complexes came to ~44,000–45,000 atoms each.

5. Simulation protocol

- **Electrostatics** — long-range Coulomb interactions were treated with **Particle-Mesh Ewald (PME)**, which computes the reciprocal-space sum efficiently and accurately for periodic systems [7].
- **Constraints and timestep** — hydrogen-bond lengths were constrained with **LINCS** [8] and **hydrogen-mass repartitioning** was applied [9], together permitting a 2 fs (up to 4 fs) timestep without loss of stability.
- **Thermostat/integrator** — temperature was controlled with a **Langevin (middle-scheme)** integrator, which gives robust and accurate configurational sampling at 300 K [10].
- **Minimize → equilibrate** — each system was energy-minimized, then equilibrated with NVT followed by NPT (Monte-Carlo barostat) at 300 K / 1 bar. Because equilibration ran on CPU for package-build tractability, a short 25 ps NVT + 25 ps NPT was used; this yields a valid production start state (positions **and** velocities **and** box preserved) but is deliberately brief.

Each equilibrated system was verified to load and integrate cleanly (finite negative potential energy ~ -6×10^5 kJ/mol, 7.65 nm cubic box) before packaging.

6. Package design

The MD package (`psoriasis_md_package.tar.gz`) ships two equilibrated RORγt systems (top hit + crystal-ligand positive control) as serialized OpenMM `system.xml` + `state.xml` + `equilibrated.pdb`, plus: `build_system.py` (rebuild any complex from receptor + ligand), `run_production.py` (production from the equilibrated state), a GPU SLURM runner, and `analyze_traj.py` (backbone RMSD, per-residue RMSF, ligand-pose stability). Production restarts directly from the NPT state — no re-equilibration.

7. Deliverables

- `psoriasis_md_package.tar.gz` — equilibrated systems, build/production/analysis scripts, SLURM runner, README.

8. Caveats

- The shipped equilibration is short (25+25 ps on CPU); for publication runs, extend it on the GPU node (a single flag change, documented in the package README).
- MD tests pose *stability*, not absolute binding free energy; rigorous affinity would require free-energy methods (FEP/TI) beyond this package's scope.

References

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